On Spurious Vorticity Generation in Vector-Coupled Metric Damping for Geophysical CFD: Theoretical Analysis of the Parasitic Curl and Cauchy Tensor Regularization

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ABSTRACT

In ultra-high-resolution mesoscale atmospheric simulations (sub-kilometer grid spacing), extreme velocity gradients across the eyewall of intense tropical cyclones induce severe grid cell deformations, frequently triggering Courant-Friedrichs-Lewy (CFL) numerical instabilities and vertical velocity (W) blow-ups. To mitigate such singularities, an intuitive approach consists in introducing a prognostic scalar metric strain variable J(\mathbf{x}, t) (governed by a transport-relaxation PDE) and applying a direct vector drag restoring force: **F**_{drag} = - K \max(0, J - 1) **u**.

In this technical note, we demonstrate analytically and numerically that this intuitive vector formulation introduces a fatal pathology: it acts as an artificial generator of vertical vorticity. By applying the curl operator to the coupled momentum equation, we establish the emergence of an anomalous, non-conservative cross-gradient term:

$\nabla \times \mathbf{F}_{drag} = - K \max(0, J - 1) \omega - K \theta(J - 1) \left(\nabla J \times \mathbf{u} \right)$

In a coherent cyclonic vortex where the metric gradient ∇J is radially inward and velocity **u** is tangential, the cross product $\nabla J \times \mathbf{u}$ aligns strictly with the vertical axis. This term injects artificial angular momentum of the same order of magnitude as the intended physical damping ($\mathcal{O}(10^{-4}\ s^{-2})$ in Category-5 Cyclone Kenneth simulations), resulting in a severe spurious spin-up (intensification exceeding 160 kt) accompanied by divergent updrafts.

We trace the root cause of this failure to the violation of Galilean invariance (material frame indifference) and Cauchy's stress theorem. Finally, we formulate the mathematically rigorous remedy: replacing the vector force with the divergence of a rank-2 symmetric stress tensor $\tau = 2\ \nu_{eff}(J)\ \mathbf{S}$, where **S** is the deviatoric rate-of-strain tensor. We formally prove the exact annihilation of the parasitic cross-product torque and the recovery of unconditionally stable enstrophy dissipation. This study provides an essential cautionary principle and design framework for geometric subgrid closures in geophysical fluid dynamics.

Keywords: Spurious vorticity, Cauchy stress theorem, Galilean invariance, Tropical cyclones, WRF, AS-Flow, Rate-of-strain tensor, Subgrid-scale instability.

1. Introduction and Problem Formulation

High-resolution numerical modeling of intense tropical cyclones (e.g., Tropical Cyclone Kenneth, 2019) at kilometer and sub-kilometer scales ($\Delta x \leq 1\text{ km}$) remains a forefront challenge in geophysical fluid dynamics. In modern compressible, non-hydrostatic models such as the Weather Research and Forecasting (WRF) model, fluid motions are integrated in terrain-following hydrostatic pressure coordinates (Skamarock and Klemp, 2008).

Near the cyclone eyewall, intense boundary layer convergence combined with violent convective updrafts generates extreme velocity strain rates:

$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

Under these localized gradients, grid cells undergo severe anisotropic elongation. When convective turnover rates surpass numerical dissipation scales, unresolved $2\Delta x$ waves accumulate at the grid cut-off scale, triggering nonlinear numerical feedback, vertical velocity divergence, and simulation failure (`NaN`).

To counteract this metric stretching, the Anti-Stretching Flow (AS-Flow) framework proposed tracking grid cell deformation via a prognostic scalar metric variable J(\mathbf{x}, t), defined such that J = 1 in the undisturbed state. In the V14 operational formulation, in order to penalize stretching (J > 1), a direct elastic drag term (termed "Kevlar force") was incorporated into the momentum tendency:

$$\boldsymbol{F}_{Kevlar} = - K_{elastic} \cdot \max(0, J - 1) \cdot \boldsymbol{u}$$

where K_{elastic} > 0 denotes a linear damping coefficient (s^{-1}) and **u** = (u, v, w)^T is the three-dimensional velocity field.

While heuristically intended to operate as an isotropic kinetic energy sink, numerical experiments revealed a stark anomaly: instead of damping turbulent kinetic energy, the addition of **F**_{Kevlar} accelerated the core winds from 100 kt to over 160 kt, producing anomalous vertical velocities exceeding 40\text{ m/s} that threatened the Courant-Friedrichs-Lewy (CFL) stability criterion.

This note provides the rigorous analytical derivation of this artifact, identifies the underlying conservation law violations, and establishes the exact tensor regularization required for physical consistency.

2. Analytical Curl Derivation and Parasitic Vorticity

Vorticity dynamics are governed by the curl of the velocity equation, ω = ∇ × **u**. To determine the torque imparted by the restoring force, we evaluate ∇ × **F**_{Kevlar}.

Expressing the threshold function via the Heaviside step function θ(x):

$$\boldsymbol{F}_{Kevlar} = - K_{elastic} \, (J - 1) \, \theta(J - 1) \, \boldsymbol{u}$$

Defining the scalar modulation factor ϕ(J) = (J - 1) θ(J - 1), we apply the vector identity for the curl of a scalar-vector product, ∇ × (ϕ **A**) = ϕ (∇ × **A**) + ∇ ϕ × **A**:

$$\nabla \times \boldsymbol{F}_{Kevlar} = - K_{elastic} \left[\phi(J) \, (\nabla \times \boldsymbol{u}) + \nabla \phi(J) \times \boldsymbol{u} \right]$$

A. Distributional Gradient Evaluation

Applying the chain rule in the sense of distributions:

$$\nabla \phi(J) = \left[\theta(J - 1) + (J - 1) \, \delta(J - 1) \right] \nabla J$$

Since the Dirac delta distribution is supported strictly at J = 1, the product (J - 1)δ(J - 1) vanishes identically. Hence:

$$\nabla \phi(J) = \theta(J - 1) \, \nabla J$$

B. Modified Vorticity Equation

Substituting back and using ∇ × **u** = ω:

$$\nabla \times \boldsymbol{F}_{Kevlar} = \underbrace{- K_{elastic} \max(0, J - 1) \, \omega}_{\text{Term I: Intended Damping}} \; \underbrace{- K_{elastic} \, \theta(J - 1) \left(\nabla J \times \boldsymbol{u} \right)}_{\text{Term II: Parasitic Vorticity Source}}$$

Theorem 1 (Spurious Curl Generation):

Any coordinate-dependent or spatially heterogeneous isotropic drag force proportional to absolute velocity ***u*** *necessarily generates an unphysical source of vorticity proportional to* ∇ *J* × ***u***.

3. Geometric Vector Analysis in Cyclonic Vortices

We analyze the orientation of Term II within a cylindrical coordinate frame (r, \lambda, z) centered on the tropical cyclone.

In a Northern Hemisphere cyclonic vortex:

1. **Primary Tangential Velocity:** The flow is dominated by cyclonic azimuthal rotation $v_\lambda > 0$:

$$\boldsymbol{u} \approx v_\lambda(r, z) \, \mathbf{e}_\lambda$$

2. **Metric Gradient Field:** Metric strain J peaks within the eyewall ($r = R_{max}$) and decreases outward into the outer vortex envelope ($r > R_{max}$). Consequently, the radial derivative is negative:

$$\frac{\partial J}{\partial r} < 0 \implies \nabla J \approx -\left| \frac{\partial J}{\partial r} \right| \mathbf{e}_r$$

Evaluating the cross product in the right-handed cylindrical basis ($\mathbf{e}_r$, $\mathbf{e}_\lambda$, $\mathbf{e}_z$) where $\mathbf{e}_r \times \mathbf{e}_\lambda = \mathbf{e}_z$:

$$\nabla J \times \boldsymbol{u} = \left(-\left| \frac{\partial J}{\partial r} \right| \mathbf{e}_r \right) \times \left(v_\lambda \, \mathbf{e}_\lambda \right) = - \left| \frac{\partial J}{\partial r} \right| v_\lambda \, \mathbf{e}_z$$

Substituting into the parasitic vorticity source term:

$$\boldsymbol{P}_z = - K_{elastic} \, \theta(J - 1) \left(\nabla J \times \boldsymbol{u} \right) = + K_{elastic} \, \theta(J - 1) \left| \frac{\partial J}{\partial r} \right| v_\lambda \, \mathbf{e}_z$$

Sign and Orientation Verdict:

Because K_{elastic} > 0, θ(J-1) ≥ 0, $|\partial_r J| > 0$, and $v_\lambda > 0$, the vertical component $\mathbf{P}_z \cdot \mathbf{e}_z$ is **strictly positive**.

In the Southern Hemisphere where $v_\lambda < 0$ and $\omega_z < 0$, $\nabla J \times \mathbf{u}$ reverses direction, making $\mathbf{P}_z$ negative and thus **systematically amplifying cyclonic rotation in both hemispheres**.

Dynamical Corollary:

The residual torque does not damp kinetic energy; it acts as a synthetic vertical vorticity pump that injects spin into the eyewall, driving the unphysical wind intensification observed in numerical simulations.

4. Scaling Analysis: Case Study of Cyclone Kenneth

To evaluate whether $\mathbf{P}_z$ is negligible or dominant, we perform a scale analysis using parameters from the WRF simulation of Cyclone Kenneth:

- Grid spacing: $\Delta x = 1000\text{ m}$
- Eyewall transition width: $\Delta L \approx 5000\text{ m}$
- Maximum tangential velocity: $U \approx 50\text{ m/s}$ (100 kt)
- Elastic restoring coefficient: $K_{elastic} = 0.05\ s^{-1}$
- Peak metric distortion: $J_{max} \approx 2.0 \implies J - 1 = 1.0$
- Core vertical vorticity: $\omega_z \approx 10^{-2}\ s^{-1}$

A. Intended Damping Rate (Term I):

$$\|\boldsymbol{D}_{stable}\| = K_{elastic} (J - 1) \omega_z \approx 0.05 \times 1.0 \times 10^{-2} = 5.0 \times 10^{-4}\ s^{-2}$$

B. Parasitic Injection Rate (Term II):

The spatial metric gradient scales as:

$$\|\nabla J\| \approx \frac{\Delta J}{\Delta L} \approx \frac{2.0 - 1.0}{5000\text{ m}} = 2.0 \times 10^{-4}\ m^{-1}$$

Yielding the vertical torque magnitude:

$$\|\boldsymbol{P}_z\| = K_{elastic} \|\nabla J\| U \approx 0.05 \times (2.0 \times 10^{-4}) \times 50 = 5.0 \times 10^{-4}\ s^{-2}$$

C. Scaling Assessment:

$$\frac{\|\boldsymbol{P}_z\|}{\|\boldsymbol{D}_{stable}\|} \approx 1.0$$

The parasitic spin-up torque equals the magnitude of the intended physical damping at $\Delta x = 1\text{ km}$. Under grid refinement ($\Delta x \to 100\text{ m}$), ΔL contracts and $\|\nabla J\| \propto 1/\Delta L$ diverges, causing the spurious source term to completely overwhelm the stable physics, driving fatal numerical blow-ups.

5. Violations of Fundamental Continuum Mechanics Principles

The failure of the vector drag formulation originates from two fundamental violations of continuum mechanics:

1. Violation of Galilean Invariance (Material Frame Indifference)

Under a Galilean boost to an inertial frame translating at constant velocity $\mathbf{V}$, the fluid velocity transforms as $\mathbf{u}' = \mathbf{u} + \mathbf{V}$.

The V14 force transforms as:

$$\boldsymbol{F}'_{Kevlar} = - K (J - 1) (\boldsymbol{u} + \boldsymbol{V}) \neq \boldsymbol{F}_{Kevlar}$$

Internal forces in a continuous medium must be strictly invariant under Galilean transformations (Noll, 1958). A fluid parcel cannot experience internal stress based on absolute velocity relative to an arbitrary numerical grid coordinate system.

2. Violation of Internal Action-Reaction Balance

In any closed fluid domain Ω, internal surface forces must integrate to zero: $\int_\Omega \mathbf{F}_{\text{int}}\, dV = 0$. A vector drag term $\mathbf{F} \propto -\mathbf{u}$ acts as an external force pulling against an unphysical stationary background, corrupting global momentum conservation.

6. Tensor Regularization via Cauchy's Stress Theorem

To restore complete physical and numerical consistency, we invoke **Cauchy's Stress Principle (1827)**:

Any internal force within a continuous medium must be expressed as the **divergence of a symmetric, rank-2 stress tensor** τ:

$$\boldsymbol{F}_{\text{internal}} = \nabla \cdot \tau, \qquad \text{with} \quad \tau = \tau^T$$

A. The AS-Flow V15 Visco-Metric Tensor

Rather than applying a direct force on **u**, the metric strain J modulates the **internal shear viscosity**:

$$\tau(J) = 2 \, \nu_{eff}(J) \, \boldsymbol{S}$$

where:

1. **S** is the deviatoric rate-of-strain tensor:
$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{1}{3} (\nabla \cdot \boldsymbol{u}) \delta_{ij}$$
2. $\nu_{eff}(J)$ is the metric-adaptive kinematic viscosity:
$$\nu_{eff}(J) = \nu_0 + \nu_{AS} \cdot \max(0, J - 1)$$

B. Proof of Parasitic Torque Annihilation

Taking the curl of the divergence of τ:

$$\nabla \times (\nabla \cdot \tau) = \nabla \times \left(\nabla \cdot [2 \nu_{eff}(J) \boldsymbol{S}] \right)$$

Using the tensor identity $\nabla \cdot \mathbf{S} = \frac{1}{2} \nabla^2 \mathbf{u} + \frac{1}{6} \nabla(\nabla \cdot \mathbf{u})$ and noting that $\nabla \times \nabla(\cdot) \equiv 0$:

$$\nabla \times (\nabla \cdot \tau) = \nu_{eff}(J) \, \nabla^2 \omega + 2 \left(\nabla \nu_{eff} \right) \cdot \nabla \boldsymbol{S} + \dots$$

Because **S** is symmetric and trace-free, **no cross-product term of the form $\nabla J \times \mathbf{u}$ can arise**. The leading-order operator on vorticity is purely parabolic:

$$\frac{\partial \omega}{\partial t} \supset \nu_{eff}(J) \, \nabla^2 \omega$$

The enstrophy $\mathcal{E} = \frac{1}{2} \int |\omega|^2 \, dV$ satisfies:

$$\frac{d\mathcal{E}}{dt} \leq - \nu_{eff}(J) \int |\nabla \omega|^2 \, dV \leq 0$$

The dissipation is unconditionally stable, preserving the integrity of the cyclonic circulation without artificial spin amplification.

7. Discussion and Recommendations for Geophysical CFD

While the tensor formulation $\tau = 2 \nu_{eff}(J) \mathbf{S}$ completely resolves the spin-up defect, an operational question arises: **does tracking an additional prognostic 3D transport PDE for J provide tangible advantages over standard atmospheric filters?**

Established mesoscale models already employ high-order horizontal diffusion (e.g., 6th-order hyperdiffusion, Skamarock and Klemp, 2008), divergence damping, and Smagorinsky subgrid closures. These filters achieve numerical stability at zero additional prognostic memory cost.

Recommended Role for Metric Invariants:

Rather than driving an active momentum feedback loop, the metric strain variable J is best repositioned as an **Adaptive Mesh Refinement (AMR) Sensor**:

1. Acting as a passive diagnostic indicator to trigger local grid nesting in dynamic shear zones.
2. Serving as a localized shock/shear detector in higher-order WENO advection schemes.

8. Conclusion

This technical note resolves the longstanding paradox of vector metric damping:

1. Direct velocity-coupled drag $-K(J-1)\mathbf{u}$ creates an inescapable parasitic vertical torque $-K(\nabla J \times \mathbf{u})$.
2. In cyclonic vortices, this term acts as an artificial spin pump, causing catastrophic unphysical intensification.

3. The only physically admissible formulation is Cauchy tensor divergence $\nabla \cdot (2\nu(J)\mathbf{S})$, which guarantees Galilean invariance and monotonic enstrophy dissipation.

Researchers developing novel subgrid closures must systematically analyze the curl of proposed forcing terms to prevent unintended vorticity generation.

Code and Data Availability

The complete standalone 2D reproduction benchmark suite (`pillar_a_spurious_vorticity_2d.py`), the WRF dynamic core Fortran routines (`module_diffusion_asflow_v14.F`), and the simulation metric datasets for Category-5 Cyclone Kenneth are openly archived on Zenodo at https://doi.org/10.5281/zenodo.placeholder and publicly accessible on GitHub at https://github.com/etonam212g5-art/AS-FLOW-RESEARCH-LAB.

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